Problem 10.37

Fund A accumulates at a force of interest

$$\frac{0.05}{1+0.05t}$$

at time $t \geq 0$. Fund B accumulates at a force of interest 0.05. You are given that the amount in Fund A at time zero is 1,000, the amount in Fund B at time zero is 500, and that the amount in Fund C at any time t is equal to the sum of the amounts in Fund A and Fund B. Fund C accumulates at force of interest δ_t. Find δ_2.

Solution

For Fund A, the accumulation factor from time 0 to time t is

$$\begin{aligned} a_A(t) &= \exp\left(\int_0^t \frac{0.05}{1+0.05x}\,dx\right) \\ &= \exp\left(\ln(1+0.05x)\big|_0^t\right) \\ &= 1+0.05t. \end{aligned}$$

Therefore, its amount function is

$$A(t) = 1{,}000(1+0.05t).$$

Fund B has the constant force of interest 0.05, so

$$B(t) = 500e^{0.05t}.$$

The amount in Fund C is the sum of the two amounts:

$$C(t) = 1{,}000(1+0.05t) + 500e^{0.05t}.$$

Differentiating,

$$C'(t) = 50 + 25e^{0.05t}.$$

The force of interest for Fund C is

$$\begin{aligned} \delta_t &= \frac{C'(t)}{C(t)} \\ &= \frac{50+25e^{0.05t}}{1{,}000(1+0.05t)+500e^{0.05t}}. \end{aligned}$$

At time $t = 2$,

$$\begin{aligned} \delta_2 &= \frac{50+25e^{0.1}}{1{,}100+500e^{0.1}} \\ &\approx \frac{77.629273}{1{,}652.585459} \\ &\approx 0.0469744. \end{aligned}$$

Thus,

$$\boxed{\delta_2 \approx 0.0469744 = 4.69744\%.}$$